```
r1.upper.middle, step 2 (2026-10-20)

 chain     seed iterations.saved acceptance.rate final.log.likelihood
     1 20263016             5000           0.277              -38595.9
     2 20263017             5000           0.277              -38592.4
     3 20263018             5000           0.276              -38602.0
     4 20263019             5000           0.276              -38589.9
 best.log.likelihood errored.proposals
            -38578.4                 0
            -38576.9                 0
            -38580.7                 0
            -38581.5                 0
                                                 file
 mcmc_r1.upper.middle_step2_chain1_2026-10-19.Rdata
 mcmc_r1.upper.middle_step2_chain2_2026-10-20.Rdata
 mcmc_r1.upper.middle_step2_chain3_2026-10-20.Rdata
 mcmc_r1.upper.middle_step2_chain4_2026-10-20.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
   age.20.to.29.transmission.multiplier.1
                                    2.146
                                  trate.1
                                    2.000
   age.30.to.39.transmission.multiplier.3
                                    1.802
               female.to.male.multiplier
                                    1.794
 age.25.to.49.hiv.mortality.multiplier.0
                                    1.664
         hiv.specific.mortality.rates.1
                                    1.602
 age.15.to.19.transmission.multiplier.0
                                    1.597
              male.awareness.multiplier
                                    1.565
 age.15.to.19.transmission.multiplier.3
                                    1.523
                                  trate.3
                                    1.480
        male.hiv.mortality.multiplier.2
                                    1.477
age.15.to.24.hiv.mortality.multiplier.1
                                    1.457
 age.20.to.29.transmission.multiplier.0
                                    1.426
         hiv.specific.mortality.rates.0
                                    1.363
        male.hiv.mortality.multiplier.1
                                    1.358
 age.30.to.39.transmission.multiplier.0
                                    1.353
 age.20.to.29.transmission.multiplier.3
                                    1.344
age.25.to.49.hiv.mortality.multiplier.1
                                    1.341
                                  trate.2
```

r1.upper.middle step 2: likelihood

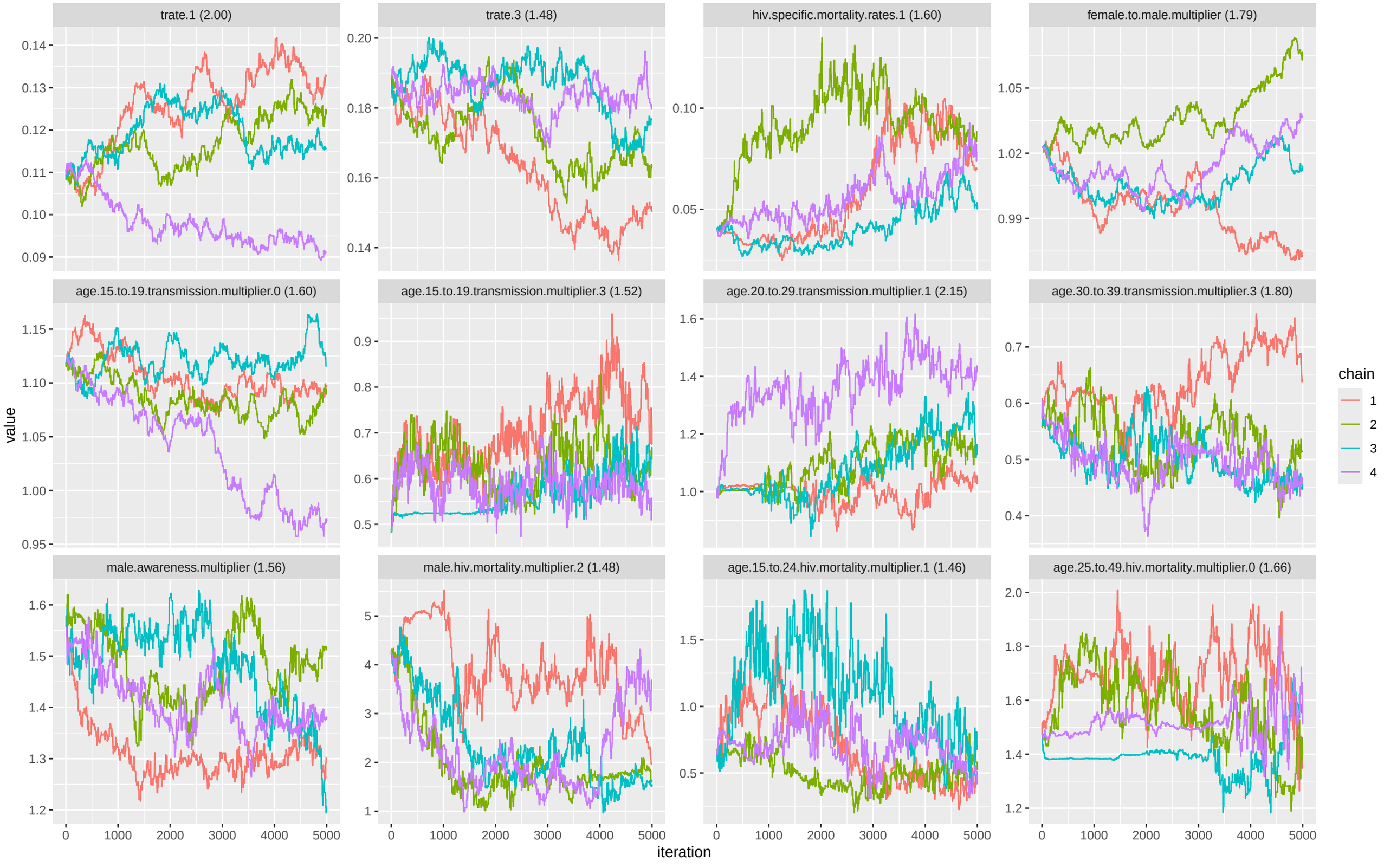
0
−10000
−20000
−30000
−40000
lue

0
25000
50000
75000
100000
iteration

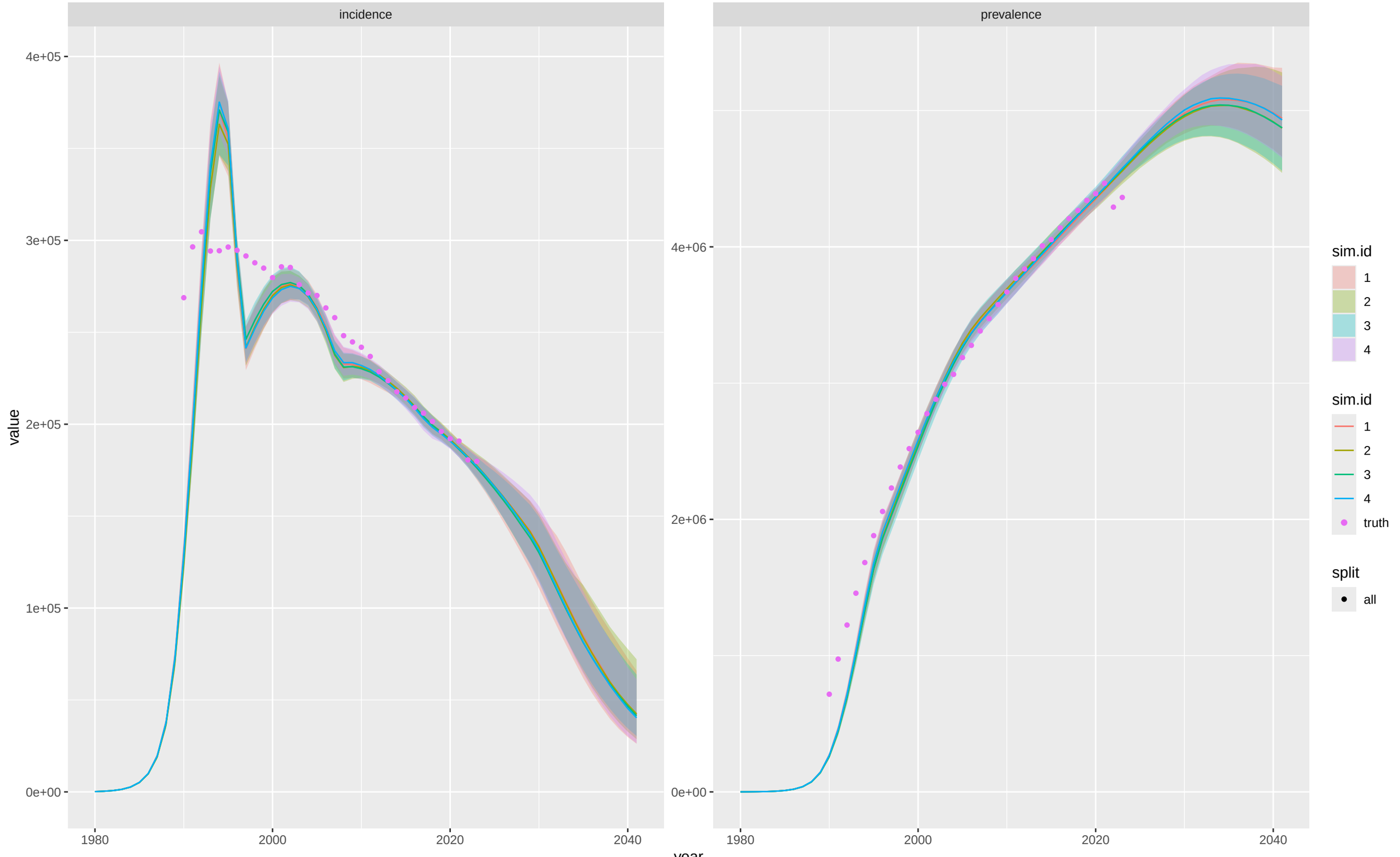
chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

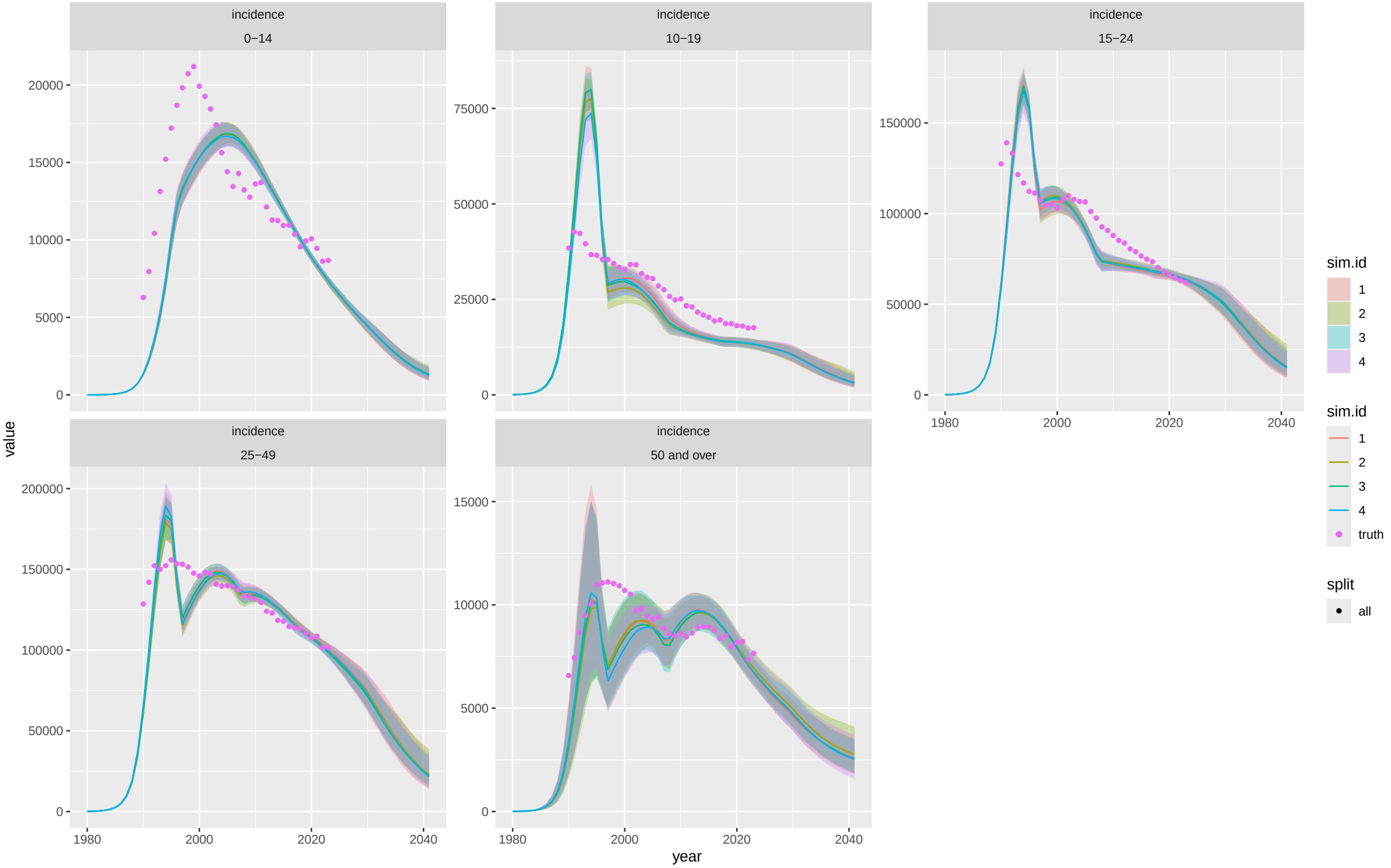
r1.upper.middle step 2: trace



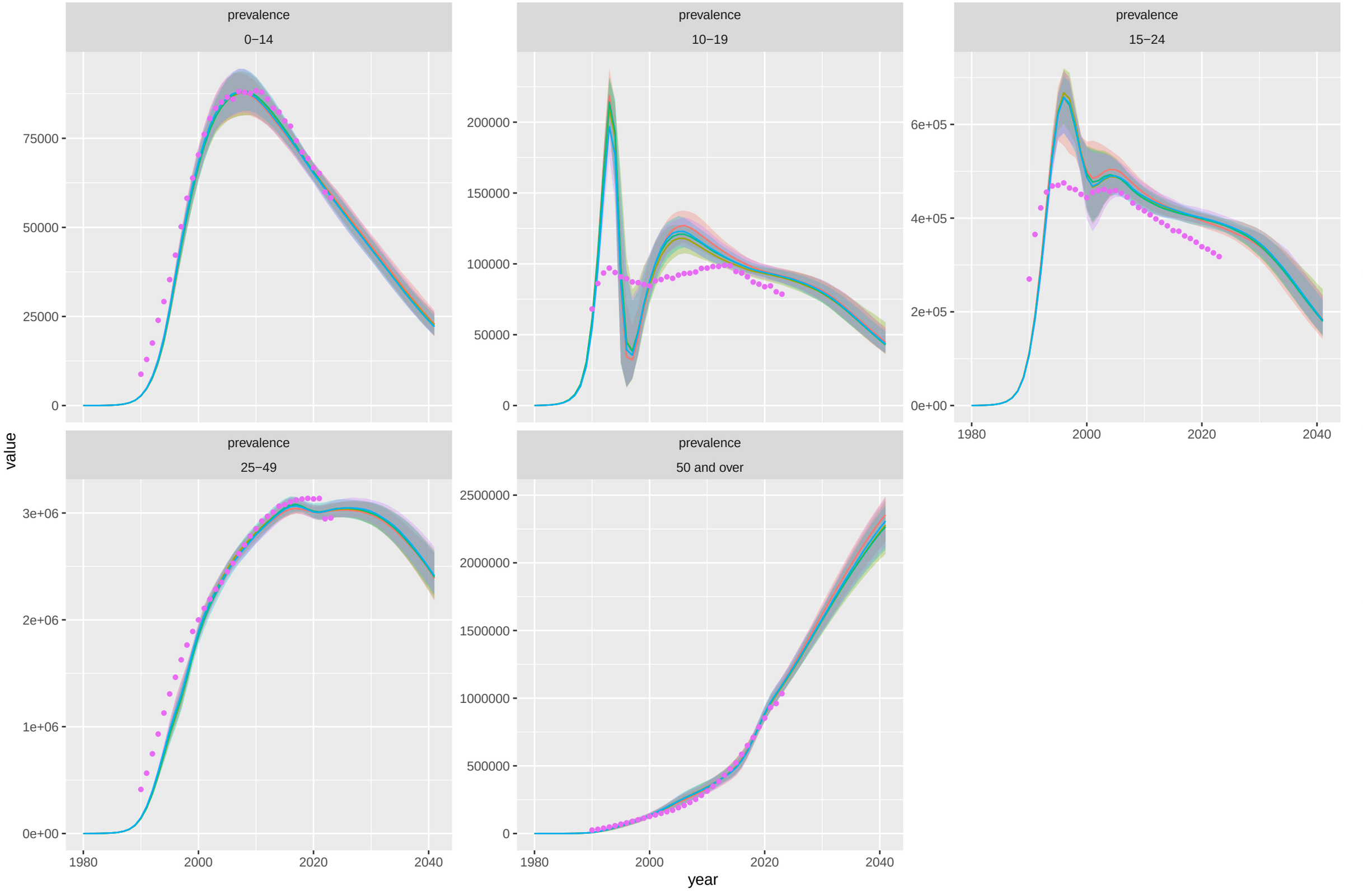
r1.upper.middle step 2: fit (one color per chain)



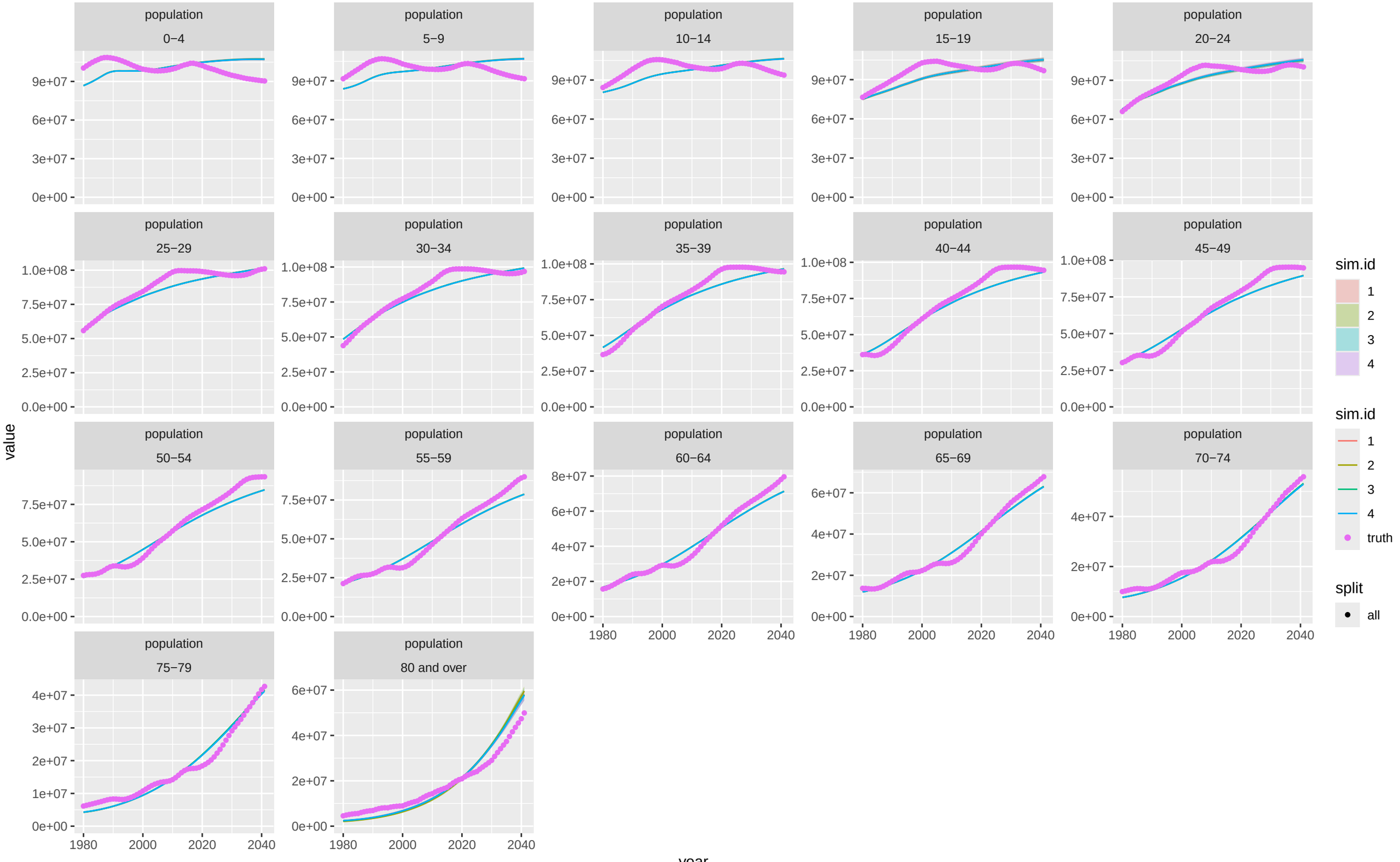
r1.upper.middle step 2: incidence.by.age (one color per chain)



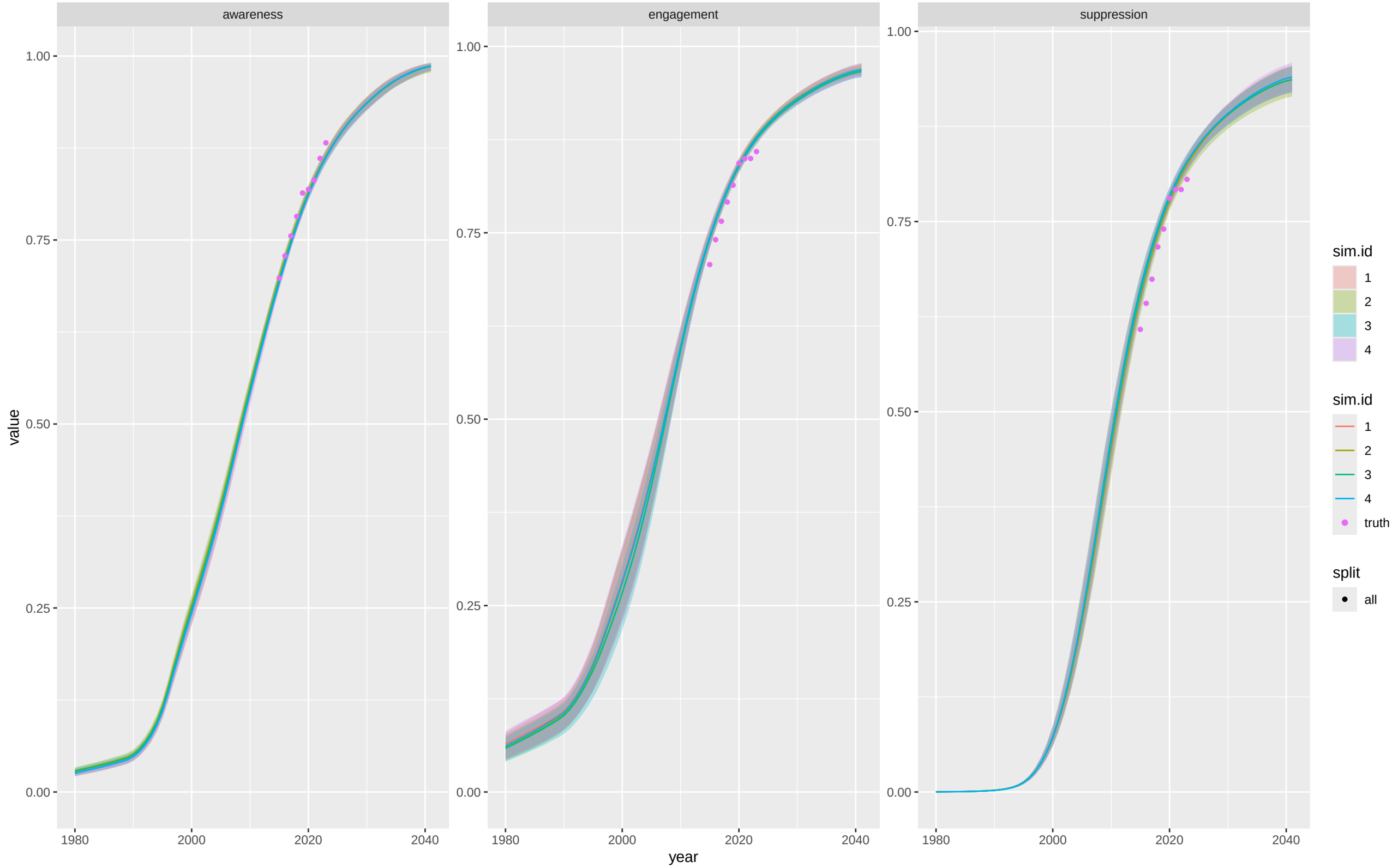
r1.upper.middle step 2: prevalence.by.age (one color per chain)



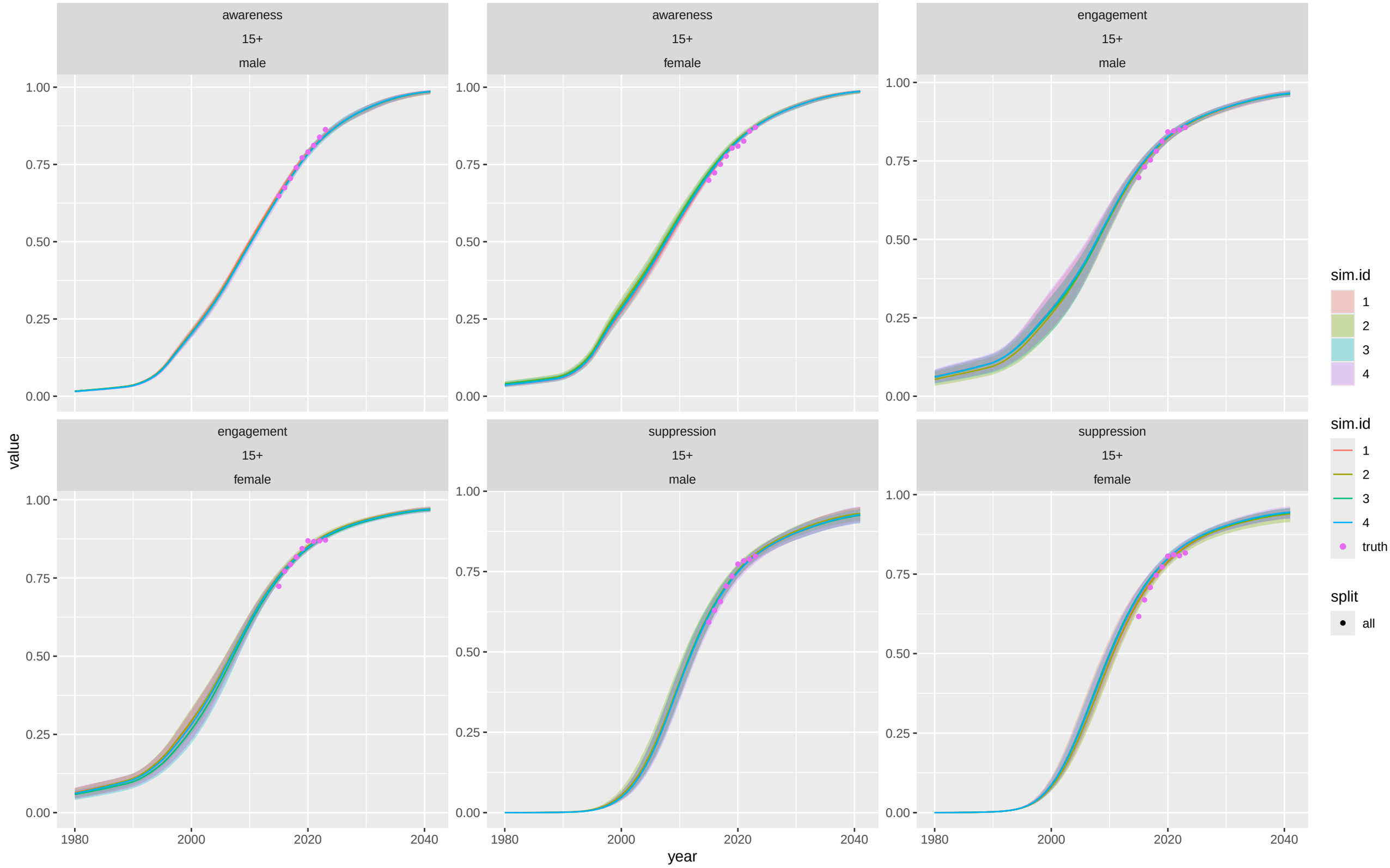
r1.upper.middle step 2: population.by.age (one color per chain)



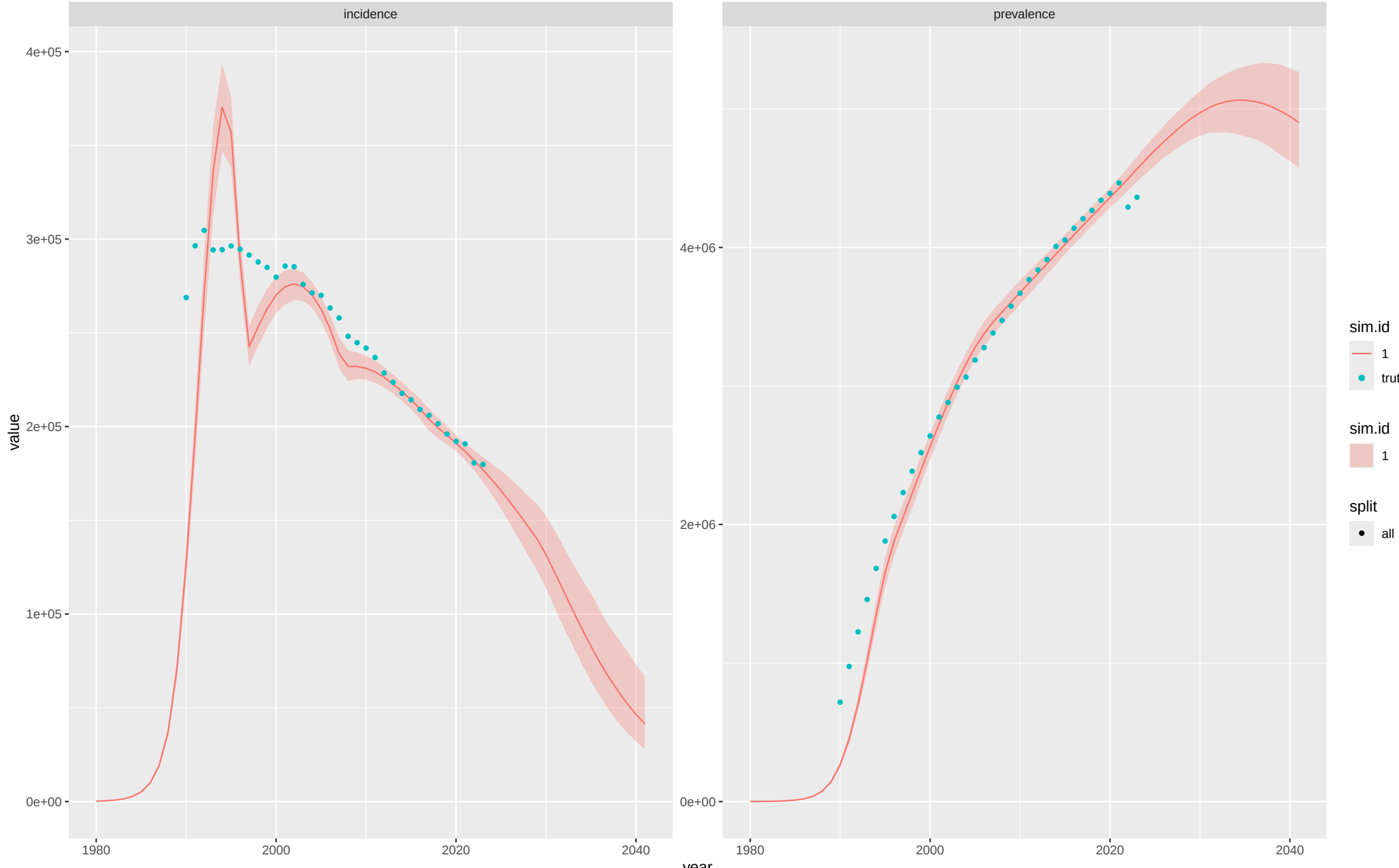
r1.upper.middle step 2: cascade (one color per chain)



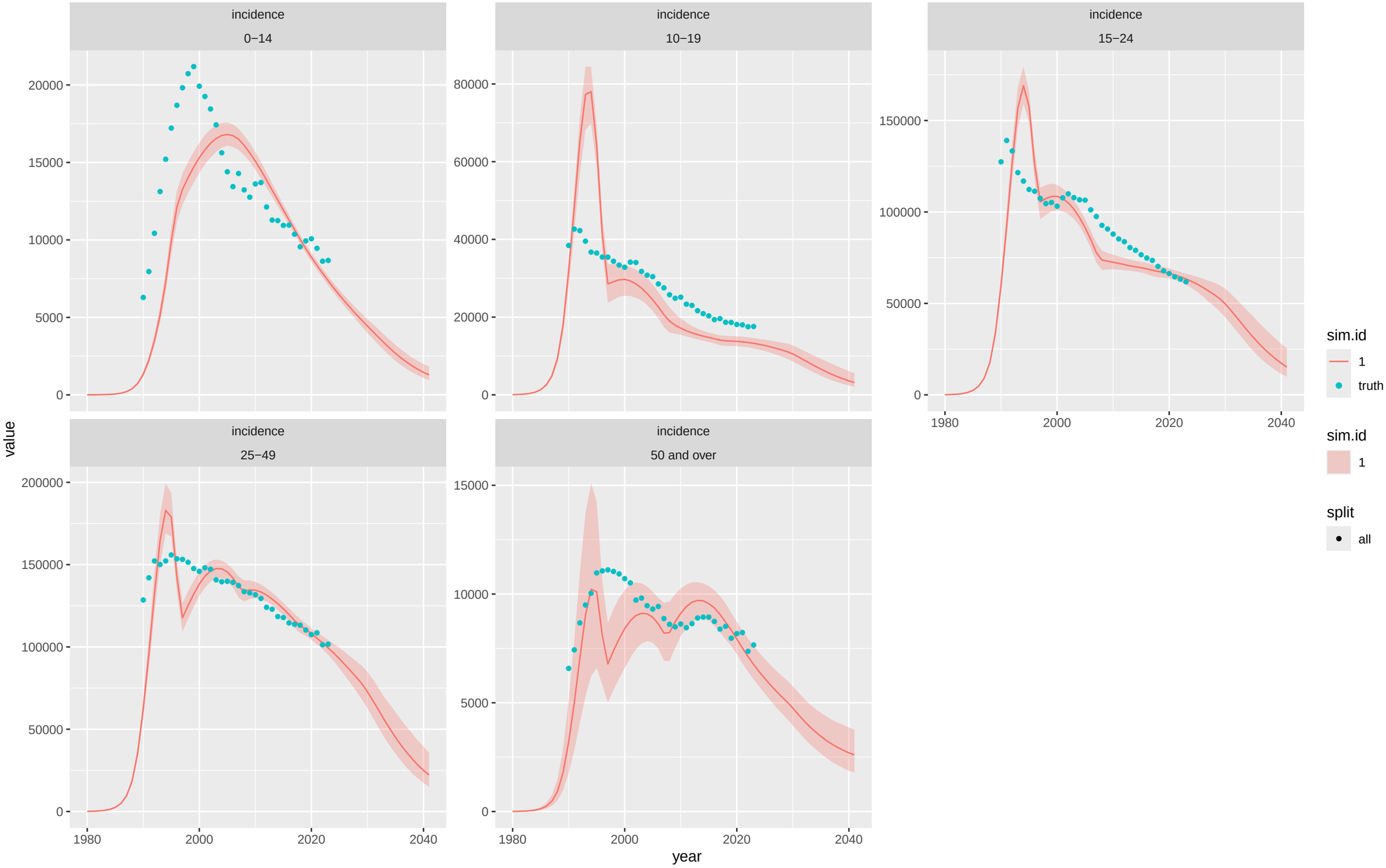
r1.upper.middle step 2: cascade.by.age.sex (one color per chain)



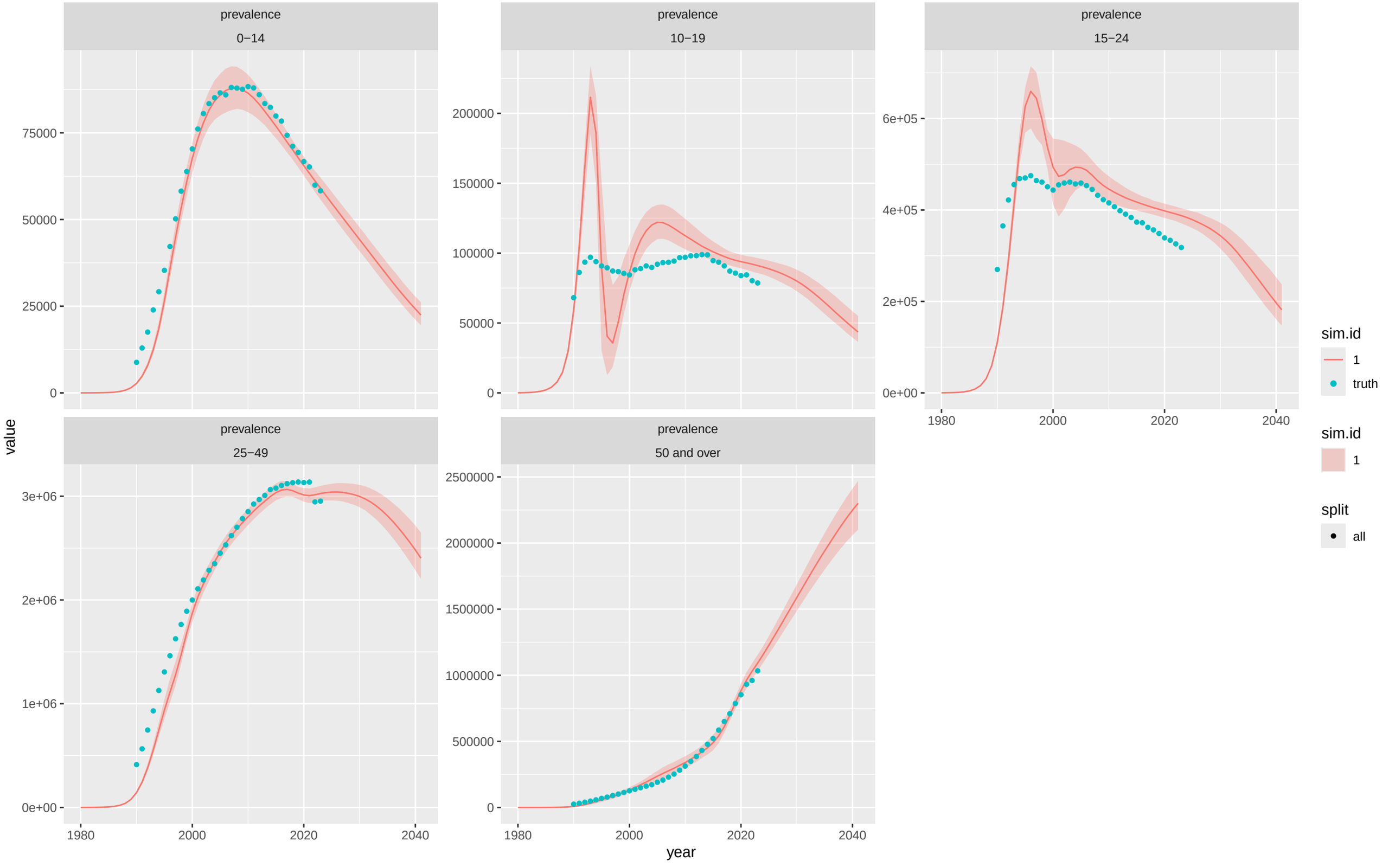
r1.upper.middle: fit (merged Step 2 simset)



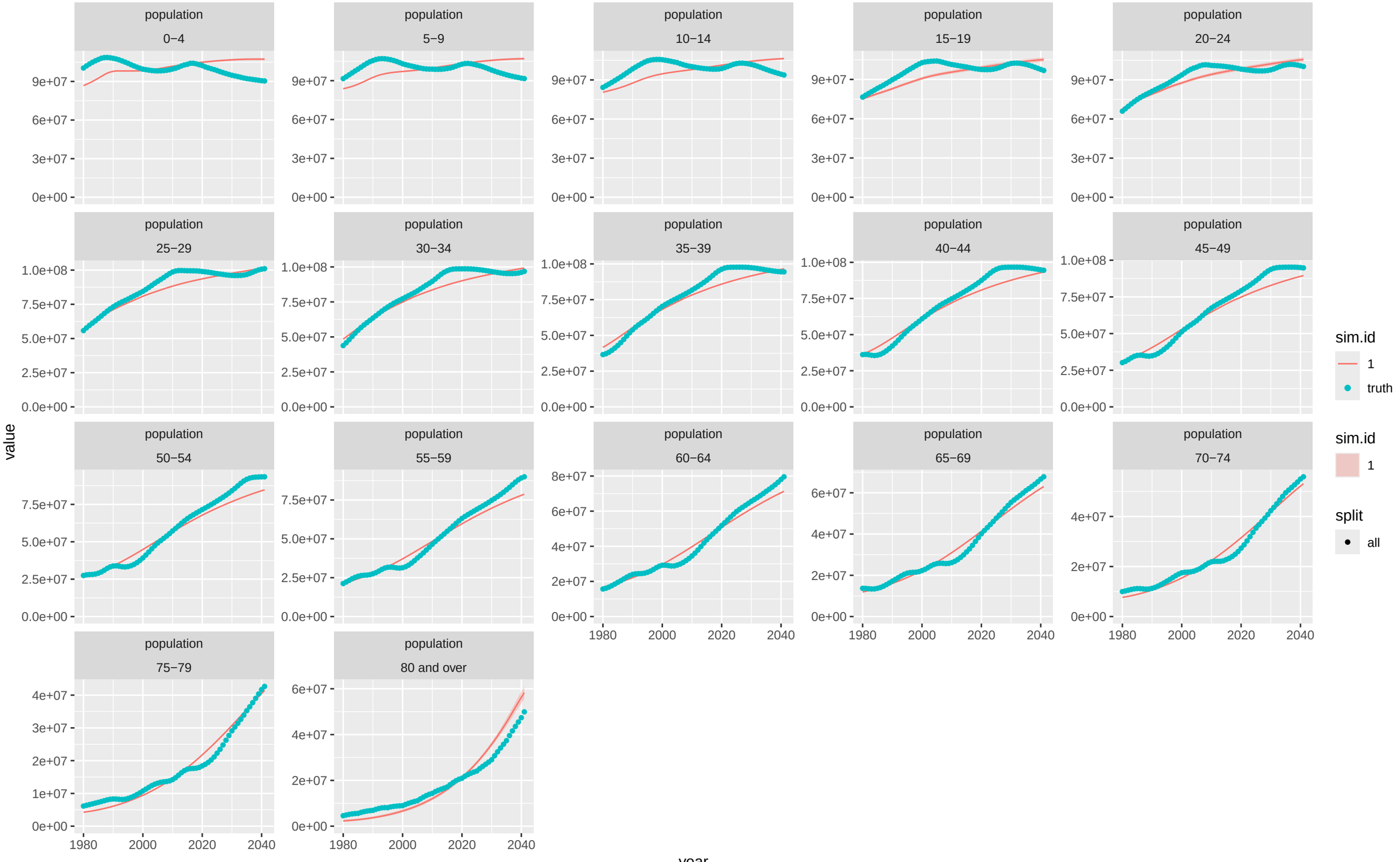
r1.upper.middle: incidence.by.age (merged Step 2 simset)



r1.upper.middle: prevalence.by.age (merged Step 2 simset)



r1.upper.middle: population.by.age (merged Step 2 simset)



r1.upper.middle: cascade (merged Step 2 simset)

awareness

engagement

suppression

value

year

1.00

0.75

0.50

0.25

0.00

1980

2000

2020

2040

sim.id

1

truth

sim.id

1

split

all

r1.upper.middle: cascade.by.age.sex (merged Step 2 simset)

